

Platelet dysfunction in vascular pathologies and how can it be treated.

Cardiovascular diseases are one of the leading causes of morbidity and mortality in industrialized countries, and although many processes play a role in the development of vascular disease, thrombosis is the primary event that precipitates stroke and acute coronary syndromes. The blood platelets are of significant importance in medicine. These cells are involved in many physiological processes, particularly haemostasis through their ability to aggregate and form clots in response to activation. In addition, these dynamic cells display activities that extend beyond thrombosis, including an important role in initiating and sustaining vascular inflammation. The expansion of knowledge from basic and clinical research has highlighted the critical position of platelets in several inflammatory diseases such as arthritis and atherosclerosis. Platelets are emerging as important mediators of inflammation and provide important signals to mediate phenotype of other blood and vascular cells. The important role of platelets in arterial thrombosis and the onset of acute myocardial infarction after atherosclerotic plaque rupture make inhibition of platelet aggregation a critical step in preventing thrombotic events associated with stroke, heart attack, and peripheral arterial thrombosis. However, the use of platelet inhibitors for thrombosis prevention must seek a delicate balance between inhibiting platelet activation and an associated increased bleeding risk. The aim of this review is to up-date the knowledge on platelets physiology and dysfunction in pathologies, such as diabetes mellitus, hypercholesterolemia, and hypertension, emphasizing the link between platelets and the inflammation-related atherosclerosis. The review evaluates the opportunities offered by the novel platelet inhibitors to efficiently alleviate the thrombotic events. Copyright © 2011 Elsevier Ltd. All rights reserved.